Symbols

$\in$	–	belongs to.
$\ni$	–	such that.
$\forall$	–	for every.
$\Rightarrow$	–	implies.
$\exists$	–	there exists

In general, the word 'operation' refers to the process of operating upon either a single or more number of elements at a time. For instance, finding the negative of an element in $\mathbb{Z}$ involves a single element at a time. So it is called an **unary operation.** On the other hand the process of finding the sum of any two elements in $\mathbb{Z}$ involves two elements at a time**.** This kind of operation is called a **binary operation** and in general an operation involving n elements is called an **n-ary operation,** $n \in \mathbb{N}$**.** In this section a detailed discussion of the binary operations is presented**.**



Learning Objectives

Upon completion of this chapter, students will be able to

- define binary operation and examine various properties
- define binary operation on Boolean matrices and verify various properties
- define binary operation on modular classes and examine various properties
- identify simple and compound statements
- define logical connectives and construct truth tables
- identify tautology, contradiction, and contingency
- establish logical equivalence and apply duality principle

12.2 Binary Operations

12.2.1 Definitions

The basic arithmetic operations on $\mathbb{R}$ are **addition** $(+)$**,** **subtraction** $(-)$, **multiplication** $(\times)$**,** and **division** $(\div)$. Eminent mathematicians of the latter part of 19thcentury and in 20thcentury like Abel, Cayley, Cauchy, and others, tried to generalize the properties satisfied by these usual arithmetic operations. To this end they developed new abstract algebraic structures through the **axiomatic approach**. This new branch of algebra dealing with these abstract algebraic structures is known as **abstract algebra.**

To begin with, consider a simple example involving the basic usual arithmetic operations addition and multiplication of any two natural numbers.

$$m+n \in \mathbb{N}\ ;\ m \times n \in \mathbb{N}\ ,\ \forall m, n \in \mathbb{N} = \{1, 2, 3, ...\}$$

Each of the above two operations yields the following observations:

(1) At a time exactly two elements of $\mathbb{N}$ are processed.

(2) The resulting element (outcome) is also an element of $\mathbb{N}$.

Any such operation defined on a nonempty set is called a **binary operation** or **a binary composition** on the set in abstract algebra.

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